

Bandwidth Characteristics of Mid-Wavelength Infrared PIN HgCdTe Avalanche Photodiodes

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Abstract—HgCdTe avalanche photodiodes (APDs) are used in mid-wavelength infrared (MWIR) photoelectric detectors in free-space optical communication, three-dimensional light detection, and ranging. Although HgCdTe has a small electronic effective mass and high mobility, there is a gap between the actual bandwidth of the device and the theoretical limit, and it is important to analyze the carrier transport mechanism in these devices. A two-dimensional finite element model is used to analyze the carrier transport mechanism for different device structures and bias voltages. The structural parameters of the device are optimized on the basis of a comprehensive determination of its bandwidth characteristics. Experimental data and calculation results are found to be consistent. This structural optimization increases the device bandwidth from 27.1 MHz to 553 MHz, with a gain-normalized dark current density (GNDCD) of less than 10^{-6} A/cm². The results obtained here can provide guidance for the design of high-speed HgCdTe devices.

Index Terms—HgCdTe avalanche photodiodes, mid-wavelength infrared, bandwidth characteristic, gain normalized dark current density

I. INTRODUCTION

During the last few years, Hg_{1-x}Cd_xTe electron avalanche photodiodes (e-APDs) have demonstrated excellent performance characteristics, including high electron saturation velocity, high quantum efficiency, and linear gain, and have become widely used in active and passive imaging, fast target detection, and single-photon detection [1-9]. A 128×128 HgCdTe APD array with a bandwidth of 100 MHz at a gain $M > 100$ has been fabricated [10]. Rothman *et al.* obtained 600 MHz bandwidth HgCdTe APDs with a cutoff wavelength of 1.55 μm [11,12]. Mitra and Beck developed a short-wavelength HgCdTe e-APD with a cutoff wavelength of 2.5 μm and achieved switching times as low as 40 μs [13]. Following illumination of the backside of a planar HgCdTe APD by a light pulse, an electron-hole pair is generated in the absorption region. When a reverse bias voltage is applied, the carriers acquire greater kinetic energy, become hot carriers, and then undergo an avalanche of collisions in the multiplication region. A large applied electric field can cause the free carrier mobility to saturate, with the device then simultaneously exhibiting characteristics of high gain and high bandwidth. However, the response characteristics of HgCdTe e-APDs devices in application are far below the theoretical limit. With existing measurement techniques, it is not possible to quantitatively determine the state of carriers when they are transported in different regions of the device, and consequently the microscopic transport mechanism of carriers within the device remains unclear. During the design of new devices, it is therefore important to analyze the influence of different factors on bandwidth.

In this work, a carrier transport mechanism for mid-wavelength infrared (MWIR) HgCdTe e-APDs is proposed on the basis of numerical simulations. The simulation results show that the acceleration of carriers in the absorption and multiplication regions is the main factor affecting device bandwidth. Taking these results into consideration, and based on existing device parameters, planar HgCdTe e-APDs with the optimized structure are fabricated by liquid-phase epitaxy (LPE). The bandwidth is measured experimentally, and it is found that for a structure-optimized device, this can reach 553 MHz, with a gain-normalized dark current density (GNDCD) less than 10^{-6} A/cm².

This work was supported by the National Key Research and Development Program of China (Grant No. 2020YFB2009300), National Natural Science Foundation of China (Grant Nos. 61725505, 62122081), Shanghai Natural Science Foundation (Grant Nos. 21ZR1473900, 21YF1454900, 21ZR1473400), Science foundation from China Postdoctoral Science Foundation (Grant No. BX2021330), Youth Innovation Promotion Association of the Chinese Academy of Sciences and Fund of SITP Innovation Foundation (Grant Nos. CX-348, CX-286). (Yue Gu and Runzhang Xie contributed equally to this work) (corresponding author: Peng Wang).

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II. MODELING FOR PHOTODIODES RESPONSE CHARACTERISTIC

To investigate the photoelectric dynamic process, a transient scan model of a HgCdTe e-APD is developed in which the bias is ramped up to the linear mode [14.15]. Once a quasi-steady state has been reached, transient scanning is started by modulating the Gaussian time behavior function. The pulse period is 2×10^{-7} s, and the photon flux remains saturated from 1×10^{-8} s to 7×10^{-8} s. The photoelectric characteristics of the device under the action of the pulse are then calculated.

In the quasi-steady-state simulation, a basic drift-diffusion transport model is considered to solve the carrier continuity equations, Poisson's equation, and the current density equation for both electrons and holes. The temperature is taken as 77 K, which reduces the dark current and ensures the device is in a state of high responsivity and high signal-to-noise ratio. The response characteristics of the device are analyzed in terms of currency fluctuations. Therefore, in the quasi-steady state simulation, Shockley-Read-Hall (SRH) recombination, tunneling, and other carrier generation-recombination mechanisms have to be included.

The SRH current is generated by defects and residual impurities from device fabrication. These become excitation or recombination centers in deep levels of the depletion region, reducing the minority carrier lifetime. In the quasi-steady state simulation, the SRH generation rate R_{SRH} is given by [16]:

$$R_{SRH} = \frac{pn - n_i^2}{\tau_p \left[n + n_i \exp\left(\frac{E_t - E_i}{kT}\right) \right] + \tau_n \left[p + n_i \exp\left(\frac{E_i - E_t}{kT}\right) \right]} \quad (1)$$

where n and p are the electron and hole densities, respectively, E_t is the position of the defect level, τ_n and τ_p are the electron and hole lifetimes, respectively, k is Boltzmann's constant, and n_i and E_i are the intrinsic carrier density and intrinsic Fermi level, respectively.

Owing to the narrow bandgap of the device, some significant tunneling currents, such as those due to band-to-band tunneling (BBT) and trap-assistant tunneling (TAT), will occur when the device operates under large reverse bias. BBT is related to the internal nonuniform local field, and the generation rate G_{BBT} and current I_{BBT} are given by [17]:

$$G_{BBT} = AF^2 \exp\left(-\frac{B}{F}\right) \quad (2)$$

$$I_{BBT} \propto \alpha (N_A)^{1/2} |V_d|^{3/2} \exp\left[-\frac{\sigma E_g^2}{(N_A |V_d|)^{1/2}}\right] \quad (3)$$

where A and B are the generation rate parameters for tunneling between bands, F is the electric field intensity, N_A is the effective doping concentration, V_d is the reverse bias, and σ

is a constant coefficient. For this n⁺-i-p device, the current is proportional to the effective doping concentration and reverse bias.

The TAT model is coupled with the SRH recombination model as follows [18]:

$$\tau_{n,p} = \frac{\tau_0}{1 + \Gamma_{n,p}} \quad (4)$$

$$\Gamma_{n,p} = \frac{\Delta E_{n,p}}{kT} \int \exp\left(\frac{\Delta E_{n,p}}{kT} u - K_{n,p} u^2\right) du \quad (5)$$

$$K_{n,p} = \frac{4}{3} \frac{\sqrt{2m_{\text{trap}}(\Delta E_{n,p})^3}}{3qh|F|} \quad (6)$$

The minority carrier lifetime is influenced by $\Gamma_{n,p}$. Here, τ_0 is an additional fitting parameter, u is the integration variable, $\Delta E_{n,p}$ is the minority tunneling energy difference, and m_{trap} is the effective quality of defects.

The carriers within the device cause avalanche impact ionization in the multiplication region. Dead space is the distance in which no impact ionization occurs. When the multiplying layer is thin, the electric field must be higher than expected to achieve a certain multiplication increment. The width of the multiplication region is narrower, making the ionization process more deterministic. In carrier avalanche generation we select the Okuto-Crowell model which is suggested by the empirical model and the dead space into account. In this simulation, the Okuto-Crowell model is adopted and generation rate [19]:

$$\alpha_{n,p}(\varepsilon) = a_{n,p} F^c \exp[-(b/F)] \quad (7)$$

$$G^{\text{Ava}} = \frac{1}{q} (\alpha_n |J_n| + \alpha_p |J_p|) \quad (8)$$

where the electron and hole collision ionization coefficients α_n and α_p , respectively, are the reciprocals of the free paths between two impacts. The coefficient $a_{n,p}$ depends on the gain curves and is inversely related to the bandgap. b and c are constant coefficients related to the depletion width and bias, and J_n and J_p are the electron and hole current density vectors, respectively.

The response time is defined as the time for the light current to increase from 10% to 90% of its maximum value. The times are taken for carriers to diffuse from absorption (t_{di}) and to transit during multiplication (t_{dr}) are the main factors determining the response characteristics. In this model, the response time is calculated as:

$$t = (t_{di}^2 + t_{dr}^2)^{1/2} \quad (9)$$

and the rise time t_r and bandwidth f_H are related by:

$$t_r f_H = 0.35 \quad (10)$$

To illustrate carrier saturation of mobility in the high electric

field for this device, the Canali model is used in the form [20]:

$$\mu(F) = \frac{2\mu_{low}}{1 + \left[1 + \left(\frac{2\mu_{low}F_{hfs}}{V_{sat}} \right)^\beta \right]^{1/\beta}} \quad (11)$$

$$V_{sat} = V_{sat,0} \left(\frac{300 \text{ K}}{T} \right)^{V_{sat,exp}} \quad (12)$$

$$F_{hfs} = \mathbf{F} \cdot \frac{\mathbf{J}_n}{|\mathbf{J}_n|} \quad (13)$$

where μ_{low} is the carrier mobility in low field. The exponent β is temperature-dependent, and F_{hfs} is the driving field. Carriers are transported from the absorption region to the electrodes by different types of driving forces [21]. When the electric field becomes large enough, the carrier velocity is no longer directly proportional to the electric field, because of scattering of carriers by defects and their local fields, and the carrier velocity saturates at a finite value V_{sat} .

III. RESULTS AND DISCUSSION

Figure 1(a) shows a cross-section of a backside-illuminated MWIR HgCdTe e-APD. Figure 1(b) shows a simplified equivalent circuit and the different mechanics of photo generated carriers in the e-APD. Table I shows the default device parameters used in the simulation. The applied electric potential shifts the energy band, which accelerates the transport of carriers toward the low electric potential. In the p-type absorption region, the photon-generated carriers are initially accelerated by the concentration gradient. Figure 1(c) shows the band diagram and carrier generation recombination velocity of the device under light irradiation. Owing to avalanche collisions, the bias voltage reaches -10 V , the carrier generation rate in the multiplication region reaches $3.34 \times 10^{23} \text{ cm}^{-3} \text{ s}^{-1}$. The maximum current density changes from 7 A/cm^2 to 11.8 A/cm^2 as it passes through the three areas of the device along the central tangent, as shown in Fig. 1(d). At a bias voltage of -10 V , the electron current density in the absorption region increases significantly, which means that the carriers have enough energy to overcome the influence of vacancies and participate in the transport process. Figure 1(e) shows the electron velocities at different voltages. After the carriers diffuse into the multiplication region, the large electric field there enables them to reach the maximum speed of the entire process. At -10 V , the carrier velocity changes from $1.1 \times 10^6 \text{ cm/s}$ (absorption region) to $2.4 \times 10^7 \text{ cm/s}$ (multiplication region), which is 1.84 times that at a voltage of -1 V . Figure 1(f) shows the cross-sectional distribution of the internal electric field of the device under pulsed-light irradiation. At -5 V , the electric field intensity in the multiplication region is only $1.7 \times 10^4 \text{ V/cm}$, but when the applied bias voltage is doubled, it reaches $3.2 \times 10^4 \text{ V/cm}$. The characteristics of device response time are all proportional to the bias voltage, and the bandwidth reaches 32.9 MHz (with a rise time of 10.62 ns) at -10 V . Although the average electric field of the multiplication region gradually increases with

increasing bias voltage, the improvement in bandwidth that can be achieved by increasing the bias voltage is still limited in the linear mode, because the carrier velocity in the multiplication region has reached saturation and is affected by ionization collisions, as shown in Fig. 1(g).

TABLE I

DEFAULT DEVICE PARAMETERS IN SIMULATIONS

Operating temperature	77 K
n ⁺ -type	$N_D = 1 \times 10^{18} \text{ cm}^{-3}$
i-type	$N_A < N_D$, $N_i = 1 \times 10^{15} \text{ cm}^{-3}$
p-type	$N_D = 1 \times 10^{16} \text{ cm}^{-3}$
Thickness	9 μm : 2 μm (n ⁺) + 3 μm (i) + 4 μm (p)
Cd composition	$x = 0.3$
Trap density	$N_t = 1 \times 10^{12} \text{ cm}^{-3}$
Light wavelength	3 μm (MWIR)
Length	50 μm
Width	20 μm
Bandgap	0.24 eV
E_{gap}	$0.25 E_g$
SRH lifetime	$\tau_n = \tau_p = 5 \text{ ns}$

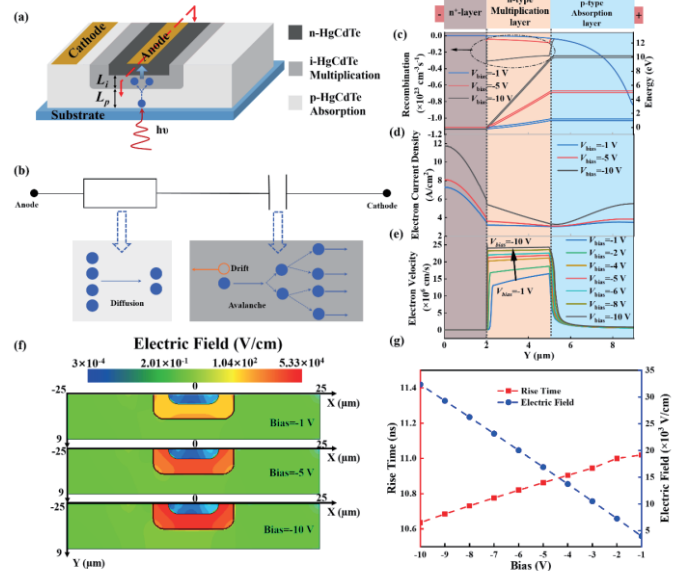


Fig. 1. (a) Cross-section of backside-illuminated n-on-p MWIR HgCdTe e-APD. (b) Simplified equivalent circuit and flow mechanics of carrier transport. (c) Simulated band diagram and carrier generation recombination velocity of the device under different biases. (d) Electron current density distribution after illumination. (e) Relationship between electron velocity and position at different biases. (f) Cross-sectional distribution of the internal electric field under pulsed-light irradiation. (g) Rise time and average electric field in the multiplication region at different biases.

After diffusing from the absorption region, the carriers begin to cross into the n-type multiplication region. Carriers moving at saturation velocity begin to collide with electrons in the multiplication region to form electron avalanches. Figure 2(a) shows the bandwidth and gain for different multiplication region thicknesses. It can be seen that the bandwidth is negatively correlated with the thickness because when carriers travel in the multiplication region at saturation velocity, shortening the travel distance allows them to be collected more rapidly by the electrode. When the multiplication layer thickness exceeds $3.5\ \mu\text{m}$, the gain of the device tends to be 3, compared with the maximum gain of 82 at $2\ \mu\text{m}$. Therefore, reducing the multiplier layer thickness can increase the bandwidth while maintaining high gain.

Figure 2(b) shows the current density, which suddenly increases when the thickness of the multiplication region is $1\ \mu\text{m}$. This is because the band-to-band tunneling current becomes dominant in the multiplication region.

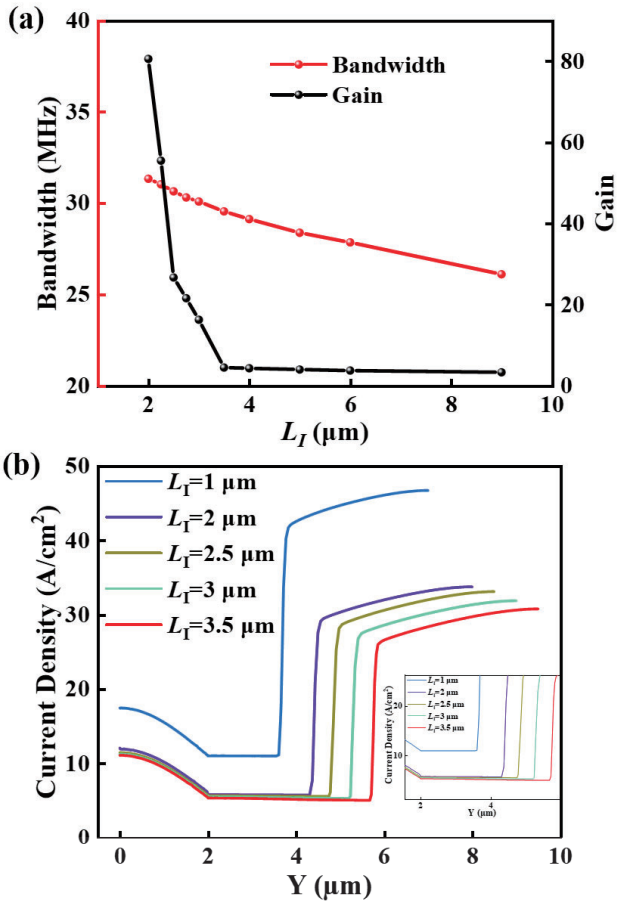


Fig. 2. (a) Effect of multiplication region thickness on bandwidth and gain. (b) The current density for different multiplication layer thicknesses.

Figure 3(a) shows the normalized response of the APD for different absorption region thicknesses. When the absorption region becomes thinner, the concentration of photogenerated

carriers' increases, the carrier diffusion due to the concentration gradient is accelerated, and the reduction in diffusion distance shortens the carrier diffusion time [22]. Reducing the thickness of the absorption region can effectively reduce the rise time of the device, which is as low as $9.953\ \text{ns}$ for a thickness of $1\ \mu\text{m}$. Figure 3(b) illustrates the rising and falling edge times of the device for different thicknesses. With the electron-dominated mechanism on which this device is based, a thin absorption region will lessen the retardation of electron motion by ionized impurity scattering. When the absorption region thickness is less than $2\ \mu\text{m}$, the response time of the device is no longer dominated by the absorption region. After the light pulse has finished, no unbalanced carriers are generated, and the device only needs to transport carriers with a certain initial velocity to the electrical contacts. The relaxation speed of the device at the falling edge will then be more strongly affected by the absorption region thickness. When this is $1\ \mu\text{m}$, the RC constant related to junction capacitance decreases the bandwidth. Figure 3(c) shows that the carrier velocity in the absorption region varies exponentially with thickness. Compared with the velocity for a thickness of $1\ \mu\text{m}$, that for $2\ \mu\text{m}$ is reduced by 43%. As the thickness of the absorption region continues to increase, the decrease in electron velocity per micrometer is no longer so great and remains at about 22%. In the multiplication region, the average electron velocity is $2.43 \times 10^7\ \text{cm/s}$ and tends to be stable. Carriers rapidly gain momentum by scattering phonons of acoustic waves and reach a large drift velocity. After a carrier is fully in the junction, its velocity tends to become stable. As the thickness decreases, the carrier velocity fluctuations become more significant.

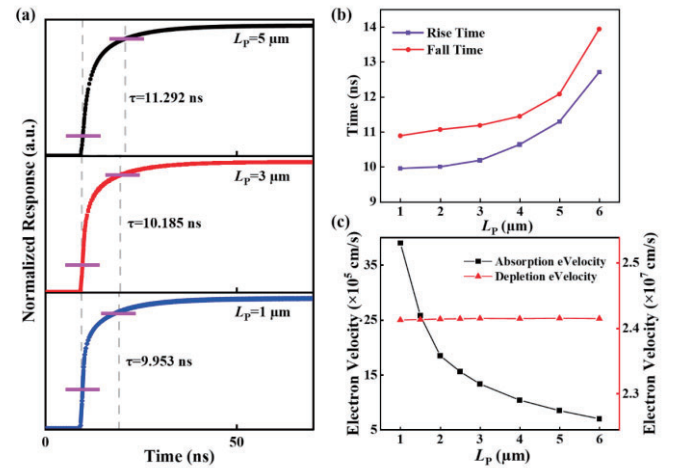


Fig. 3. (a) Response time of devices with different absorption region thicknesses. (b) Relationship between extracted response time and absorption thickness. (c) Simulated carrier velocity in absorption and depletion regions.

Figure 4(a) shows the simulated bandwidth characteristics of the device for different electrode spacings and an electrode length of $1\ \mu\text{m}$. When the electrode spacing is $24\ \mu\text{m}$, the device bandwidth is only $42\ \text{MHz}$, but it increases by a factor of 5.4 when the spacing is reduced to $9\ \mu\text{m}$. It can be seen that reducing the electrode spacing helps to reduce the transport

time of carriers after transit. Figure 4(b) shows the space charge distribution in the device after pulsed-light irradiation. Thinning the absorption region increases the space charge concentration and the lateral expansion, thus reducing the time required for the lateral transport of carriers. The space charge is $2.13 \times 10^{10} \text{ cm}^{-3}$ at a thickness of $1 \mu\text{m}$, the carrier transport path is relatively fixed, and the motion to the P electrode is faster in a thin absorption region.

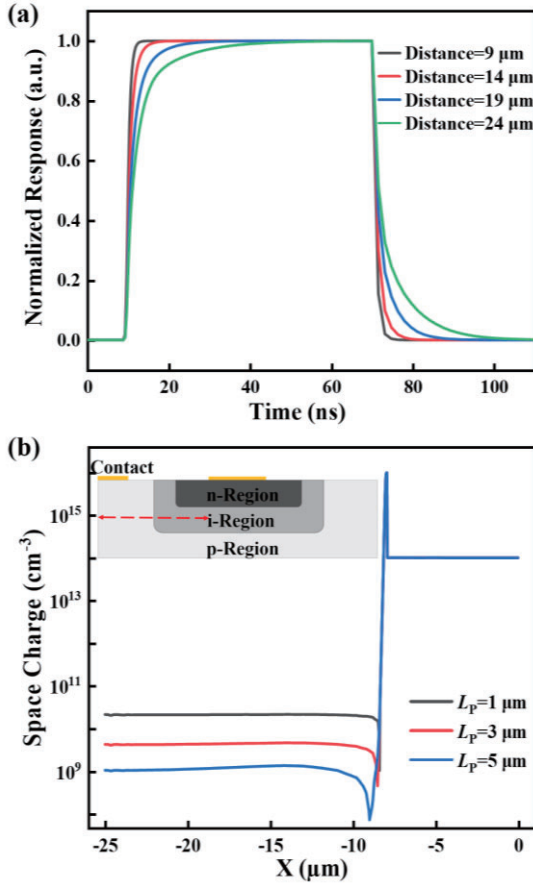


Fig. 4. (a) Bandwidth characteristics of the device for different electrode spacings. (b) Lateral distribution of space charge of electrons in the device.

IV. EXPERIMENTAL VERIFICATION

Figure 5(a) shows the response measurement system. The pulsed beam is generated by a distributed feedback laser (DFB) and becomes a hybrid frequency band laser under the LiNbO₃ modulator. The device under test is placed in a Dewar at 77 K to ensure a stable test temperature. Bias-T provides DC voltage to the device. The radio-frequency signal is amplified by a low-noise amplifier (LNA) and then enters the digital storage oscilloscope (DSO) and vector network analyzer (VNA) (Keysight N5222B). Figure 5(b) shows a photograph of the device. B ions of energy 150 keV are implanted from

the surface, generating a large number of Hg interstitials to occupy the intrinsic Hg vacancies and form an n-type injection region. Meanwhile, the particle injection generates a large number of defects inside the material, and rapid annealing in a nitrogen atmosphere causes these free donors to diffuse into the absorption region. This process makes the junction area deeper, with some vacancies being occupied and exhibiting donor characteristics [23,24]. There is a donor-dominated low-concentration region ($1 \times 10^{15} \text{ cm}^{-3}$), and the diffusion depth of Hg interstitials is about $3 \mu\text{m}$ [25]. The performance of the device is closely related to the concentration and thickness of its doped region. The accuracy of the simulation is dependent on the actual structural parameters of the device and is crucial to the subsequent structural optimization procedure. It is difficult to simulate the annealing process after ion implantation, and so the quality of the junction has to be evaluated by the differential Hall method. The quality of the results is limited by the accuracy of this method, and so the estimate of the lower carrier concentration may not be accurate enough [26], as shown in Fig. 5(c). The bandwidth response characteristics of the device at 77 K are measured by changing the bias, and the device bandwidth is found to be 27 MHz at -10 V , and in the simulations the characteristics of device response time are all proportional to the bias voltage, as shown in Fig. 5(d). Figure 5(e) shows the redesigned device bandwidth, which reaches 553 MHz with a GNDCCD $< 10^{-6} \text{ A/cm}^2$, achieved with high gain and low dark current. The current characteristics of the device are also shown. And the simulation results were found to be in accordance with the experimental data. [27]. A comparison of performance before and after optimization shows that the proposed model demonstrates good agreement with experimental data.

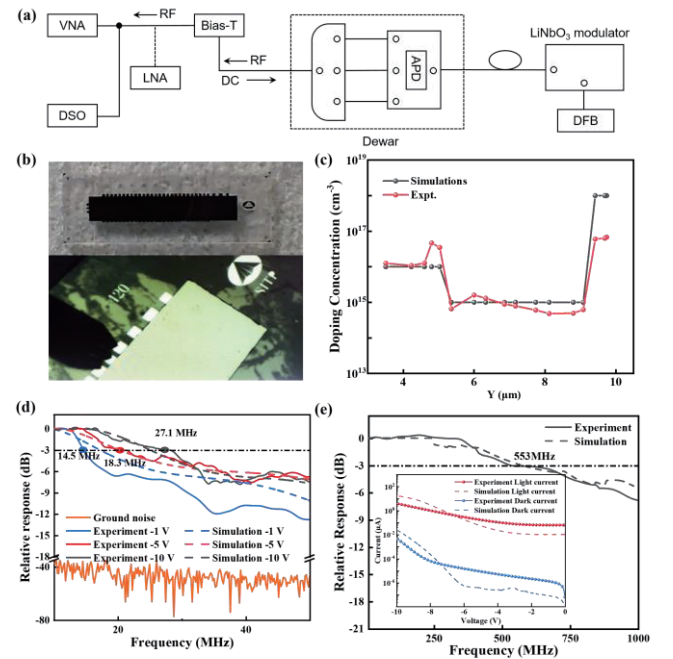


Fig. 5. (a) Experimental system for high-speed response device. (b) Photograph of experimental sample. (c) Spatial distribution of doping concentration measured by

differential Hall method and from simulation. (d) Device frequency response before structural optimization under different bias conditions for simulations and experiments. (e) The simulations and experiments for optimized device frequency response and current characteristics.

V. CONCLUSION

In summary, numerical models for investigating the bandwidth and carrier transport mechanism of an MWIR HgCdTe e-APD have been established. The effects of the bias, absorption region, and multiplication region have been studied. The results show that the absorption region thickness and lateral transport of carriers within the device are the most significant factors. The bandwidth increases by 13% when the absorber thickness decreases from 4 μm to 1 μm . A reduction in the device electrode spacing from 24 μm to 9 μm gives a 228 MHz bandwidth, which is 5.4 times that previously achieved. Owing to carrier generation-recombination and scattering in the multiplication process, a change in the multiplication region thickness has no significant effect on bandwidth. Since the carrier velocity reaches saturation, the multiplication region should be thick enough to reduce tunneling currents. In the multiplication region, the carriers reach saturation velocity, ensuring a balance between bandwidth and absorption efficiency. Structural optimization of the device has been performed. The maximum bandwidth reaches 553 MHz, with $\text{GND CD} < 10^{-6} \text{ A/cm}^2$. The results of simulations and experiments presented here can provide a basis for device design and the development of novel high-bandwidth devices.

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